
On the Out-of-Sample Predictability of Short-Horizon Cryptocurrency Returns

Mehmet Demir Guven
Department of Computer Science
ETH Zürich

Abstract

We ask whether daily cryptocurrency returns can be forecast out of sample well enough to survive transaction costs, and we design the study so that the question is falsifiable rather than flattering. Three assets, two horizons, and six forecasters are evaluated by purged and embargoed walk-forward cross-validation over 2020–2026, giving 18 machine-learning settings measured against a martingale null. The conclusion turns on a choice of test that the applied literature usually makes by default. Diebold–Mariano rejects in favour of the random walk in 7 of 18 settings and never in favour of a model, which would suggest that learning actively hurts. But the random walk is *nested* inside every model considered, and under nesting that statistic is biased toward the smaller model. Under the Clark–West correction, 5 of 18 settings reject the no-predictability null at an uncorrected 5% level, against 0.9 expected from a search of that size; 2 survive control of the false discovery rate and none survives control of the family-wise error rate. The survivors are not usable forecasts: their out-of-sample R_{OS}^2 against a recursive drift benchmark is -0.0079 and $+0.0031$, and their net Sharpe intervals straddle zero. No setting exhibits significant sign-timing skill, which is the property a directional strategy actually trades. The settings whose Sharpe intervals exclude zero are disjoint from those surviving statistical correction, and every high-Sharpe result in the study is long-only exposure in disguise. We report a faint, fragile statistical signal and no tradeable one, together with the full apparatus that would have detected either.

1 Introduction

Claims of accurate cryptocurrency price prediction are abundant and rarely survive scrutiny. The reported accuracy is usually traceable to one of four errors, each of which is well documented and each of which is easy to commit without noticing.

Target leakage. The predicted quantity is a function of contemporaneous or future information, so the model fits an identity rather than forecasting [1]. Leakage of this kind is now understood to be a leading cause of irreproducible findings in machine-learning-based science more broadly [2].

Preprocessing leakage. A scaler, imputer, or feature selector is fit on the full sample before splitting, so the test distribution informs training.

Absent benchmarks. Accuracy is reported on a price series rather than on returns. Prices are close to a martingale, so a naive random-walk forecast attains near-perfect R^2 ; a model that appears to explain 99% of price variance may explain none of the return variance, which is the only component that can be traded [3].

Silent multiple testing. Many model, asset, and horizon combinations are tried and the best is reported without adjusting the threshold. In the closely analogous cross-sectional asset-pricing literature, Harvey et al. [4] argue that conventional significance levels are badly miscalibrated once the size of the collective search is acknowledged.

This paper is organised so that each of these is either impossible by construction or measured and reported. The contribution is not a model. It is an evaluation protocol applied honestly to a question with a commercially attractive answer, which returns the unattractive one.

A second contribution is methodological, and it changed our own conclusion. The standard tool for comparing forecast accuracy, the Diebold–Mariano test [5], assumes the competing forecasts are non-nested. The comparison that the return-predictability literature actually makes—a conditional model against a random walk—violates that assumption, because setting the model’s slopes to zero recovers the benchmark exactly. Under the null the larger model must still estimate coefficients that are truly zero, and the resulting estimation noise inflates its sample mean squared prediction error even though its population error is identical. Section 4.8 sets this out, and Section 5.2 shows that on our data the correction moves the verdict from “nothing predicts, several are significantly worse” to “five settings reject, two survive false-discovery control, none is tradeable.” Both statements are defensible from the same forecasts. Only the second is correct.

2 Related work

Return predictability and out-of-sample evaluation. Welch and Goyal [6] showed that a long list of equity-premium predictors that perform well in sample fail to beat a simple historical mean out of sample, and Campbell and Thompson [7] formalised the out-of-sample R^2 against that benchmark which we adopt in Section 4.6. The lesson generalises: in-sample fit is close to uninformative about forecast value, and the appropriate benchmark is a naive one estimated recursively rather than with hindsight.

Machine learning in asset pricing. Gu et al. [8] conduct a large-scale comparison of machine-learning methods for equity returns and find measurable but small out-of-sample gains, concentrated in methods that shrink aggressively. Our model roster is chosen in that spirit: regularised linear models and a deliberately constrained gradient booster, on the premise that an unregularised learner in this signal-to-noise regime fits noise and a null result obtained from an overfit model would be uninformative.

Backtest overfitting. Bailey et al. [9] and Arnott et al. [10] document how repeated backtesting manufactures apparently significant strategies, and Bailey and López de Prado [11] propose the deflated Sharpe ratio, which raises the bar in proportion to the number of trials. White [12] and Sullivan et al. [13] address the same problem for technical trading rules through a bootstrap reality check. We adopt the deflated Sharpe ratio together with explicit control of the false discovery rate and the family-wise error rate over our grid.

Cryptocurrency market efficiency. The empirical record is mixed and appears to be time-varying. Urquhart [14] reports evidence of inefficiency in early Bitcoin returns; Nadarajah and Chu [15] find that a simple power transformation of the same returns satisfies efficiency tests, and Tiwari et al. [16] report that Bitcoin is largely efficient over most of their sample. This literature evaluates statistical efficiency; it generally does not ask whether any detected dependence survives transaction costs, which is the question we take up.

Validation for dependent data. Purged and embargoed cross-validation is due to López de Prado [17], and is the mechanism by which we prevent a training label whose realisation overlaps a test window from informing the fit.

3 Data

We use daily adjusted OHLCV bars from Yahoo Finance for the USD pairs of BTC, ETH, and SOL. BTC and ETH span 2019-01-01 to 2026-07-18 (2,756 bars each); SOL begins 2020-04-10 (2,291

Table 1: Evaluated out-of-sample window per setting. Counts are forecasts, not trades; the trading rule of Section 4.5 samples every h -th bar.

Asset	h (days)	OOS forecasts	First	Last
BTC	1	2186	2020-07-23	2026-07-17
BTC	7	2174	2020-07-29	2026-07-11
ETH	1	2186	2020-07-23	2026-07-17
ETH	7	2174	2020-07-29	2026-07-11
SOL	1	1721	2021-10-31	2026-07-17
SOL	7	1709	2021-11-06	2026-07-11

bars). Cryptocurrency markets trade every calendar day, so the bar index is contiguous with no weekend or holiday gaps. We verified this rather than assuming it, and Section 4.3 shows that the split logic remains conservative under gaps regardless.

The evaluated out-of-sample window is shorter than the data window because the first fold consumes 504 bars of training history. Table 1 gives the resulting per-setting counts.

Asset selection is not survivorship-free. BTC, ETH, and SOL were chosen in knowledge of their survival, while many contemporaneous tokens did not survive. This biases the study toward finding profitable long exposure, which strengthens rather than weakens a negative result about *predictability*, but would invalidate any cross-sectional claim. Section 7 returns to it.

4 Method

4.1 Features

Let C_t, H_t, L_t, V_t denote close, high, low, and volume at bar t . Every feature is a function of information available at or before t , and all twelve are scale-free so that a single model can pool across assets trading at very different price levels. We use cumulative log returns over $k \in \{1, 5, 10, 21\}$, realised volatility over $w \in \{10, 21\}$, Wilder’s RSI over 14 bars recentred to roughly $[-1, 1]$, a price-normalised MACD histogram, two moving-average ratios, a mean normalised range, and a volume z -score:

$$r_t^{(k)} = \log \frac{C_t}{C_{t-k}}, \quad \sigma_t^{(w)} = \text{sd} \left(r_{t-w+1}^{(1)}, \dots, r_t^{(1)} \right), \quad (1)$$

$$\text{RSI}_t = 100 - \frac{100}{1 + \bar{U}_t / \bar{D}_t}, \quad \text{macd}_t = \frac{M_t - S_t}{C_t}, \quad (2)$$

$$\text{ma}_t = \log \frac{\text{SMA}_7(C)_t}{\text{SMA}_{21}(C)_t}, \quad \text{dist}_t = \log \frac{C_t}{\text{SMA}_{50}(C)_t}, \quad (3)$$

$$\text{range}_t = \frac{1}{14} \sum_{i=0}^{13} \frac{H_{t-i} - L_{t-i}}{C_{t-i}}, \quad z_t = \frac{\log(1 + V_t) - \mu_t^{(21)}}{s_t^{(21)}}, \quad (4)$$

where $\bar{U}$ and $\bar{D}$ are exponentially smoothed gains and losses with $\alpha = 1/14$, and $M_t = \text{EMA}_{12}(C)_t - \text{EMA}_{26}(C)_t$ with $S_t = \text{EMA}_9(M)_t$. The longest warm-up is 50 bars; rows with any undefined feature are dropped.

The absence of look-ahead is enforced by a test rather than by inspection: future bars are perturbed and every feature value at or before t is asserted unchanged. This catches an accidental centred window, a sign error in a shift, or a non-causal smoother, none of which is reliably visible when reading code.

4.2 Prediction target

The label attached to a decision made at t is the return realised over $(t, t + h]$:

$$y_t^{(h)} = \log \frac{C_{t+h}}{C_t}. \quad (5)$$

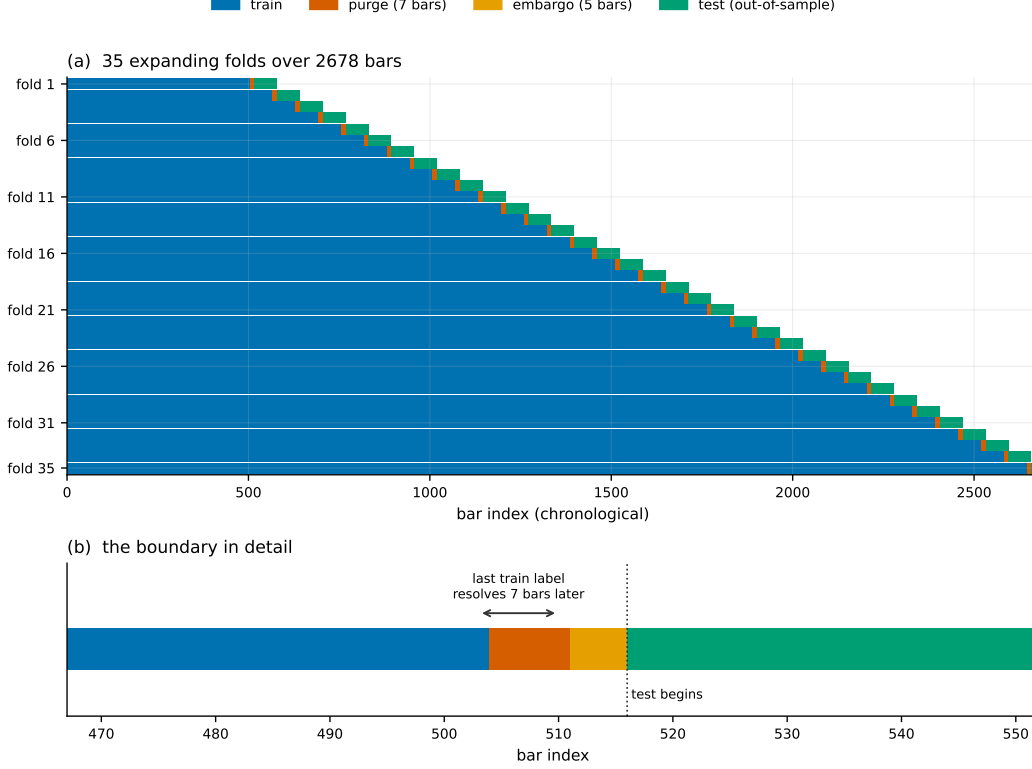


Figure 1: The validation design. Panel (a) shows all 35 folds on one timeline; panel (b) zooms on a single train/test boundary, where the purge and the embargo are individually visible. Equation (6) is what the gap enforces.

The final h rows have no label. We retain them with features and a missing target, because that is exactly the state a live forecaster occupies on the most recent bar, and we exclude them from fitting and scoring.

4.3 Purged, embargoed walk-forward validation

Ordinary k -fold cross-validation shuffles time and trains on the future. A plain chronological split is also insufficient: because $y_t^{(h)}$ resolves at $t + h$, a training row close to the test block carries a label whose realisation overlaps the test window. Following López de Prado [17], we purge the h rows before each test block and impose a further embargo of e bars. Writing $\mathcal{T}_k$ and $\mathcal{V}_k$ for the train and test index sets of fold k , every fold satisfies

$$\max(\mathcal{T}_k) + h + e < \min(\mathcal{V}_k). \quad (6)$$

Test blocks tile forward without overlap. We use an expanding window with $|\mathcal{T}| \geq 504$ bars, $|\mathcal{V}| = 63$ bars, and $e = 5$, giving 35 folds for BTC and ETH and 28 for SOL. Figure 1 shows the resulting layout, drawn from the splits the study actually runs.

Purging is applied by row position, whereas the leak it prevents is a calendar one. Position-based purging remains valid under missing bars, and conservatively so: gaps only stretch the timeline, so h positions always span at least h calendar days. We assert this on a synthetic index with a quarter of its bars removed.

Two further controls are worth naming because they are commonly missed. Standardisation is a pipeline step fit inside each fold, so it never sees test rows. And the gradient booster purges its own early-stopping holdout: without a gap, the last h rows of its fitting slice carry labels realised inside that holdout, so early stopping would be tuned on partly observed outcomes. We apply the same h -bar purge one level down, and test it by corrupting exactly those rows and asserting the fitted model is unchanged.

4.4 Forecasters

Three benchmarks and three machine-learning models are refit from scratch on every fold. The benchmarks are the random walk $\hat{y}_t = 0$, the training-window mean $\hat{y}_t = \bar{y}_T$, and an AR(1) regression of $y^{(h)}$ on $r^{(1)}$. The machine-learning models are ridge ($\alpha = 1$), elastic net ($\alpha = 10^{-3}$, $\rho = 0.5$), and XGBoost [18] with depth 3, learning rate 0.03, subsampling 0.8, $\lambda = 1$, and early stopping on the purged temporal holdout described above. The linear objectives are

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2, \quad \hat{\beta}_{\text{EN}} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \rho \|\beta\|_1 + \frac{\alpha(1-\rho)}{2} \|\beta\|_2^2. \quad (7)$$

Every one of the three collapses to the random walk when its slopes are zero. They are nested extensions of the benchmark, which Section 4.8 shows is not a detail of presentation.

4.5 Trading rule and transaction costs

Forecasts become unit positions on a non-overlapping schedule: one decision every h bars, held to the next. Sampling every h -th bar prevents the same h -day return being counted h times. Costs are charged on turnover, so reversing a position pays for two sides:

$$w_t = \text{sign}(\hat{y}_t) \in \{-1, 0, +1\}, \quad \tau_t = |w_t - w_{t-1}|, \quad \pi_t = w_t \left(e^{y_t^{(h)}} - 1 \right) - \tau_t c, \quad (8)$$

with $c = 17$ basis points per side (10 bp taker fee, 5 bp slippage, 2 bp half-spread). Equity compounds as $E_T = \prod_{t \leq T} (1 + \pi_t)$ from unity.

Because decisions are taken every h bars, the Sharpe ratio annualises at $365/h$ rather than at 365. Using the latter inflates a seven-day strategy's Sharpe by a factor of $\sqrt{7} \approx 2.6$.

The schedule admits h start offsets and nothing distinguishes them. We therefore run every strategy on all h and report the spread, which Section 5.6 shows is not a formality.

An always-long reference is computed on the same schedule with the same cost model, so that a forecaster whose only achievement is holding the asset can be recognised as such.

4.6 Out-of-sample R_{OS}^2

We report the Campbell–Thompson form [7], whose denominator is the squared error of a genuine ex-ante benchmark forecast rather than of the realised mean of the evaluation window:

$$R_{OS}^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \hat{y}_t^b)^2}, \quad (9)$$

with $\hat{y}^b$ the recursively estimated historical mean. That benchmark scores exactly zero against itself, which serves as a wiring check on the metric. The alternative denominator $\sum_t (y_t - \bar{y})^2$ uses a quantity unknowable in advance and flatters the benchmark; we compute it as a cross-check but do not quote it.

4.7 Diebold–Mariano and the nested-model problem

For loss differential $d_t = L(e_t^m) - L(e_t^b)$ under squared-error loss, the Diebold–Mariano statistic [5] is $DM = \bar{d} / \sqrt{\hat{V}}/n$ with $\hat{V}$ a Newey–West long-run variance [19] truncated at $h - 1$ Bartlett lags, appropriate because h -step forecast errors follow an MA($h - 1$):

$$\hat{V} = \hat{\gamma}_0 + 2 \sum_{k=1}^{h-1} \left(1 - \frac{k}{h}\right) \hat{\gamma}_k, \quad \hat{\gamma}_k = \frac{1}{n} \sum_{t=k+1}^n (d_t - \bar{d})(d_{t-k} - \bar{d}). \quad (10)$$

The $1/n$ normalisation applies to every autocovariance including the lagged ones. This is deliberate: the Bartlett weights guarantee $\hat{V} \geq 0$ only under that normalisation, and the more intuitive $1/(n - k)$ can drive a weighted sum negative in small samples. We apply the Harvey–Leybourne–Newbold small-sample correction [20], $DM^* = DM \sqrt{(n+1-2h+h(h-1)/n)/n}$, against a t_{n-1} reference. A negative statistic favours the model, and the test is two-sided because the null is equal accuracy and a model can be significantly worse.

4.8 Clark–West

Because the random walk is nested in each of our models, Diebold–Mariano is undersized as a test of predictability: under the null the larger model pays sample mean squared prediction error to estimate coefficients that are truly zero, and the test reads that cost as evidence for the benchmark. Clark and West [21] subtract the term explicitly. With $\hat{y}^b$ the nested benchmark and $\hat{y}^m$ the larger model,

$$\hat{f}_t = (y_t - \hat{y}_t^b)^2 - \left[(y_t - \hat{y}_t^m)^2 - (\hat{y}_t^b - \hat{y}_t^m)^2 \right], \quad (11)$$

and $\bar{f}/\sqrt{\hat{V}_f/n}$ is referred to a standard normal, one-sided. A *positive* statistic favours the model, opposite to Diebold–Mariano, which is reason enough never to report either p -value without its statistic.

We verify the bias by simulation rather than asserting it. Over 200 replications in which the larger model forecasts pure noise, so that the null holds by construction, the mean Diebold–Mariano statistic is positive and it rejects in favour of the benchmark far above the nominal rate, while Clark–West remains centred on zero and holds its size.

4.9 Sign timing

A model can be poor at magnitude and useful at direction, which is all a sign strategy requires. Pesaran and Timmermann [22] test independence between the sign of the forecast and the sign of the outcome. With P the hit rate, P_y and P_x the proportions of positive outcomes and forecasts, and $P_* = P_y P_x + (1 - P_y)(1 - P_x)$,

$$S = \frac{P - P_*}{\sqrt{\widehat{\text{var}}(P) - \widehat{\text{var}}(P_*)}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \widehat{\text{var}}(P) = \frac{P_*(1 - P_*)}{n}, \quad (12)$$

$$\widehat{\text{var}}(P_*) = \frac{(2P_y - 1)^2 P_x(1 - P_x)}{n} + \frac{(2P_x - 1)^2 P_y(1 - P_y)}{n} + \frac{4P_y P_x(1 - P_y)(1 - P_x)}{n^2}. \quad (13)$$

A positive statistic indicates sign-timing skill and a negative one indicates a reliably wrong-way forecast. The statistic is undefined when the forecast never changes sign, which is precisely the case for the always-long drift models, and we report it as missing rather than as a number.

4.10 Sharpe ratio, deflation, and interval estimates

Sharpe ratios are noisy in short samples [23] and inflated by selection. The probabilistic Sharpe ratio [11] gives the probability that the true Sharpe exceeds a benchmark SR^* , correcting for sample length, skewness γ_3 , and kurtosis γ_4 :

$$\widehat{\text{PSR}}(\text{SR}^*) = \Phi \left[\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4} \widehat{\text{SR}}^2}} \right]. \quad (14)$$

The deflated Sharpe ratio sets SR^* to the expected maximum across N trials, so the bar rises with the size of the search:

$$\mathbb{E} \left[\max_{i \leq N} \text{SR}_i \right] \approx \sigma_{\text{SR}} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{N\epsilon} \right) \right], \quad (15)$$

with γ the Euler–Mascheroni constant. Here N is the number of models raced within one asset–horizon, which is a floor on the true search; the reported deflated Sharpe is therefore an upper bound and we label it as such. Confidence intervals come from a circular block bootstrap [24] with block length $\lfloor n^{1/3} \rfloor$ and 500 resamples, reported as percentile intervals.

4.11 Multiple testing

Eighteen settings tested at the 5% level produce $0.05 \times 18 = 0.9$ rejections from noise alone. We therefore report raw counts alongside two adjustments of the Clark–West p -values over the family

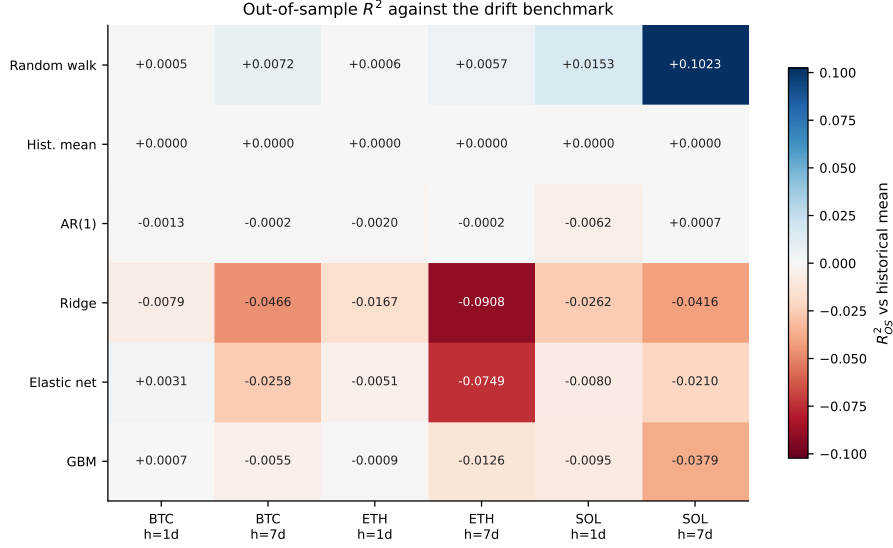


Figure 2: Out-of-sample R^2_{OS} against the drift benchmark. Red is worse than an ex-ante historical-mean forecast, blue better. The benchmark row is exactly zero by construction, which fixes the scale.

of 18. Holm–Bonferroni [25] controls the family-wise error rate and Benjamini–Hochberg [26] the false discovery rate:

$$p_{(i)}^{\text{Holm}} = \max_{j \leq i} \min\{(m - j + 1)p_{(j)}, 1\}, \quad p_{(i)}^{\text{BH}} = \min_{j \geq i} \min\left\{\frac{m}{j} p_{(j)}, 1\right\}. \quad (16)$$

Benchmarks are excluded from the family: they are the null being tested against, not candidates in the search.

5 Results

5.1 Point accuracy

Out-of-sample R^2_{OS} is negative in 16 of 18 machine-learning settings, from -0.0908 (ETH, $h = 7$, ridge) to $+0.0031$ (BTC, $h = 1$, elastic net). Rank information coefficients lie between -0.080 and $+0.053$. Figure 2 gives the full surface and Table 3 the underlying numbers.

That ridge attains $CW = +2.62$ with $p = 0.004$ on BTC at $h = 1$ while its R^2_{OS} is -0.0079 is not a contradiction. Clark–West asks whether the *population* mean squared prediction error is lower, and a model can satisfy that while its own parameter-estimation noise leaves the realised forecast worse than a constant. It is evidence of a weak signal, not of a usable forecast.

5.2 Predictive accuracy against the random walk

The two tests disagree, and the disagreement is the finding. Table 2 summarises. Under Diebold–Mariano the conclusion would be that machine learning is actively harmful here: no setting beats the random walk and seven lose to it significantly. Under the test valid for nested models, five settings reject the null of no predictability. Both numbers derive from identical forecasts.

Ordered by p -value the five are BTC ridge ($p = 0.004$), BTC elastic net (0.005), BTC GBM (0.022), and ETH ridge (0.031), all at $h = 1$, together with BTC elastic net at $h = 7$ (0.039). Four of the five are BTC and four are at the one-day horizon. That concentration is consistent with a weak short-horizon effect in the most liquid asset, and equally consistent with chance over a search of this size.

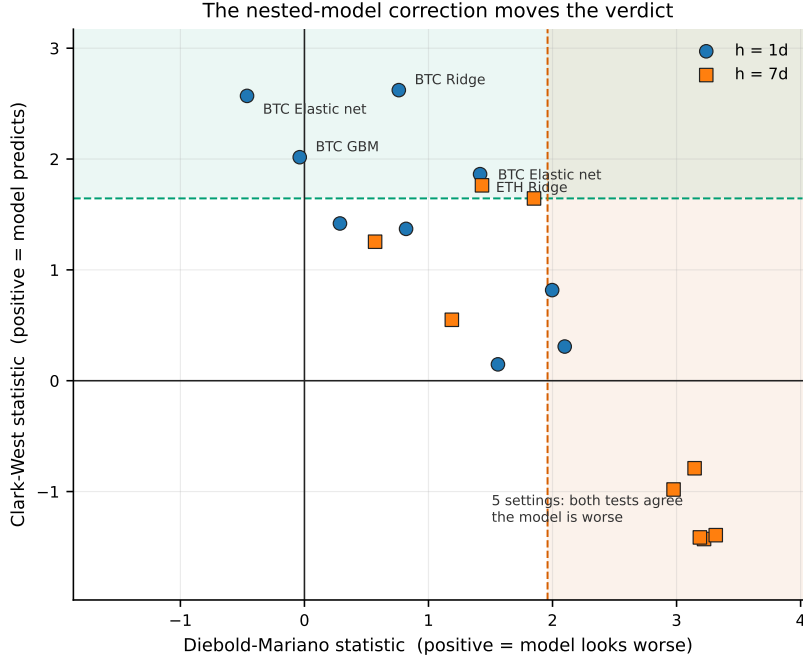


Figure 3: Every setting placed by its Diebold–Mariano statistic against its Clark–West statistic. The shaded vertical band marks where DM calls the model significantly worse; the horizontal band marks where Clark–West rejects no-predictability. Points falling in both are the settings DM misreads. The two tests disagree by construction under nesting, and the scatter shows the disagreement rather than asserting it.

Table 2: Outcome of each test and correction across the 18 machine-learning settings. Direction is required, not merely significance: a Diebold–Mariano rejection is attributed to whichever side the statistic’s sign favours.

Criterion	Settings
Diebold–Mariano, favours model	0 / 18
Diebold–Mariano, favours benchmark	7 / 18
Clark–West, uncorrected	5 / 18
Clark–West, Benjamini–Hochberg	2 / 18
Clark–West, Holm–Bonferroni	0 / 18
Pesaran–Timmermann, sign skill	0 / 18
Pesaran–Timmermann, anti-predictive	2 / 18
Net Sharpe interval excludes zero	2 / 18
<i>Expected by chance at $\alpha = 0.05$</i>	<i>0.9 / 18</i>

5.3 Multiple testing

Controlling the false discovery rate at 5% leaves BTC ridge and BTC elastic net at $h = 1$, both with $p^{\text{BH}} = 0.046$, marginally under the line. Controlling the family-wise error rate leaves nothing: the smallest raw p -value, 0.0044, does not clear the Holm threshold of $0.05/18 = 0.0028$ (Figure 4). The evidence for predictability is real but marginal, and would not survive a stricter reader.

5.4 Sign timing

Directional accuracy spans 0.471 to 0.528 and every setting lies inside its coin-flip band (Figure 5). No setting exhibits significant sign-timing skill. Two are significantly *anti*-predictive: ETH ridge and ETH elastic net at $h = 7$ give $S = -3.29$ and $S = -3.33$, both $p \approx 0.001$, with hit rates of 0.476 and 0.479.

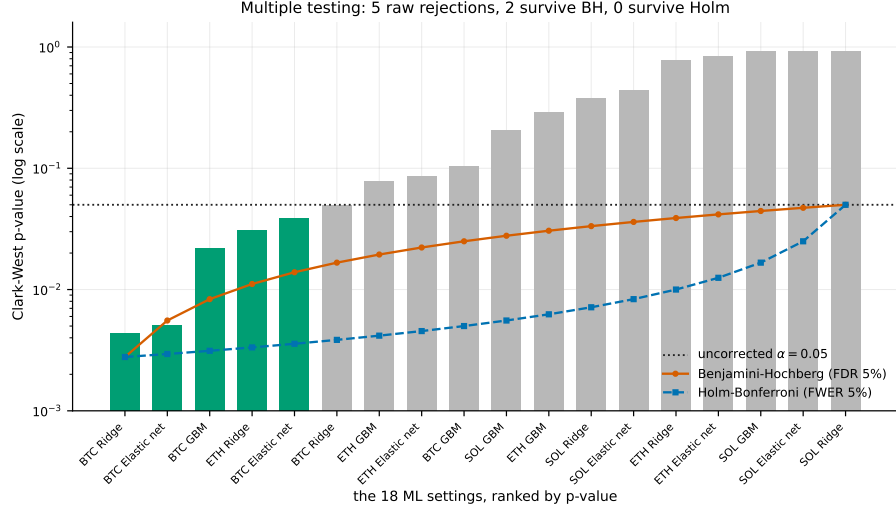


Figure 4: The 18 Clark–West p -values ranked against the Holm–Bonferroni and Benjamini–Hochberg thresholds, on a logarithmic scale because the entire decision occurs below $p = 0.05$. A bar must fall below a line to be rejected by that procedure. Testing 18 settings at the 5% level is expected to yield 0.9 rejections from noise alone.

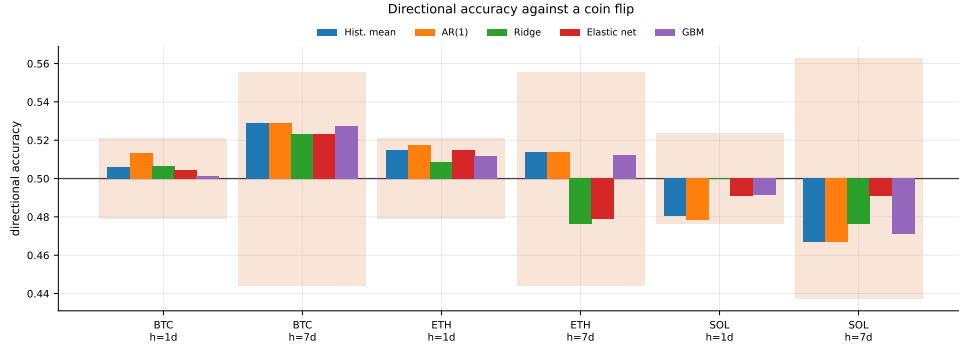


Figure 5: Directional accuracy against the band a fair coin would occupy. The band uses the effective sample size n/h : consecutive h -day forecasts are built from overlapping windows, and treating each as independent would shrink the band by $\sqrt{h}$ and manufacture significance.

Those two illustrate why we never report a bare p -value. Presented as “ $p = 0.001$ ” alone they read as the strongest findings in the study. They are among the worst.

5.5 Economic performance

Two of 18 settings have a net Sharpe whose interval excludes zero, both gradient boosting on ETH: 0.91 with interval $[0.00, 1.74]$ at $h = 1$, and 0.92 with $[0.02, 1.80]$ at $h = 7$. Both intervals begin essentially at zero. Neither setting appears among the two surviving Benjamini–Hochberg correction, and neither of those two has an interval excluding zero.

The statistical winners and the economic winners are therefore disjoint. Two unrelated groups of survivors drawn from one search is the signature of noise: if a genuine short-horizon effect existed and were tradeable, the settings that detect it and the settings that profit from it ought to overlap.

Six of 18 settings out-Sharpe buy-and-hold on the point estimate, the weakest form of that claim given that every interval contains zero. Meanwhile the historical-mean forecaster attains Sharpe ratios of 0.84, 0.80, and 0.25 across the three assets, which is impressive until one observes that it makes exactly one position change. It predicts a positive drift, hence is always long, and its net

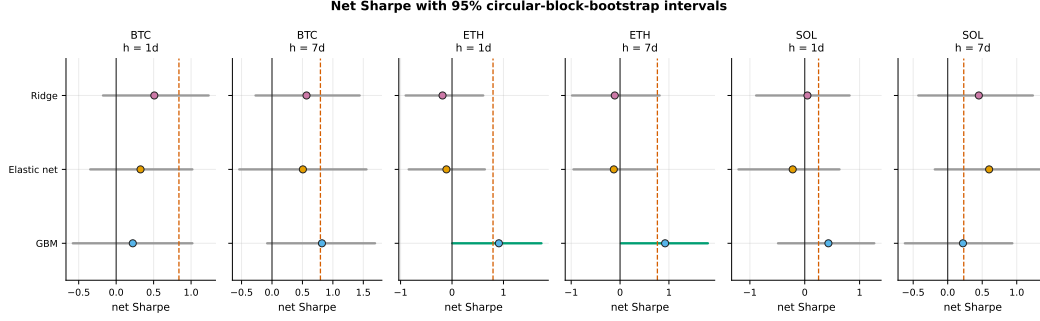


Figure 6: Net Sharpe with 95% circular-block-bootstrap intervals. The dashed line marks buy-and-hold on the same schedule and cost model. Intervals excluding zero are highlighted.

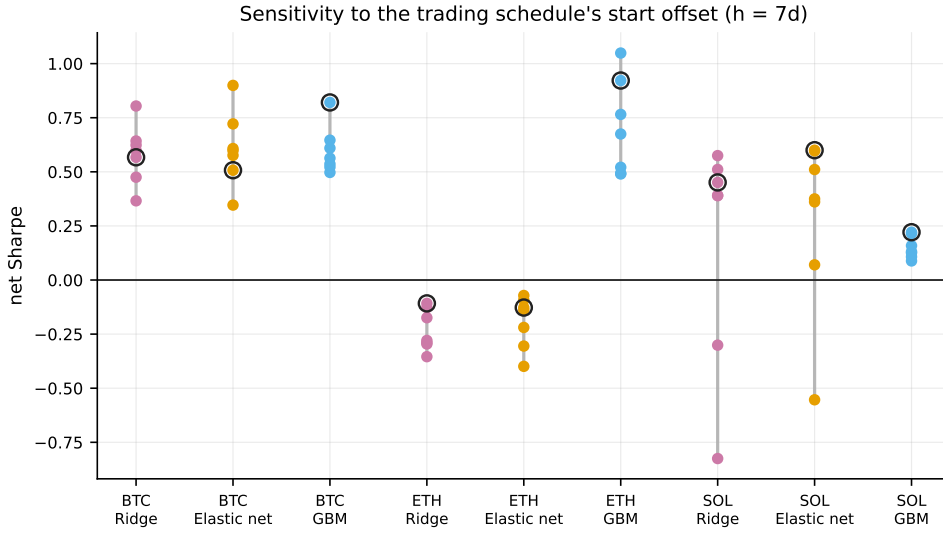


Figure 7: Net Sharpe on each of the h possible start offsets of the trading schedule at $h = 7$. The circled point is the offset the results table reports; nothing privileges it.

returns are numerically identical to buy-and-hold asset by asset and horizon by horizon. We carry buy-and-hold as an explicit row in Table 4 so that this cannot be mistaken for model performance.

Maximum drawdowns range from -55% to -99% across the machine-learning settings. This is a property of the position sizing rather than of any model: unit leverage reversing direction on roughly 70% annualised volatility carries a large variance drag, so the geometric return sits well below the arithmetic one. Buy-and-hold itself draws down between 76% and 96% over the same window.

5.6 Robustness to the trading schedule

Sampling every h -th bar leaves h schedules and nothing privileges the one starting at the first bar. At $h = 7$ the spread is substantial: SOL ridge reports a Sharpe of 0.45 drawn from a distribution running from -0.83 to 0.57 , and SOL elastic net reports 0.60 from a range of $[-0.55, 0.60]$. The headline number is in part a property of where sampling happened to begin. A study reporting a single-offset Sharpe at a multi-day horizon without this check is reporting one draw and calling it an estimate.

6 Discussion

Three points seem worth carrying forward.

The choice of test changed the answer. Diebold–Mariano is the default in applied forecast comparison, and it is the wrong default whenever the benchmark is nested, which it is in nearly every return-predictability study that pits a model against a random walk. The difference here is not cosmetic: “0 of 18, and 7 significantly worse” versus “5 of 18 reject.” A reader shown only the first would conclude the models are harmful; a reader shown only the second, without the multiple-testing correction, would conclude something was found. Neither is the result.

Statistical significance and economic value separated cleanly. The settings that reject no-predictability have negative or near-zero out-of-sample R^2_{OS} and Sharpe intervals straddling zero, while the settings that make money do not reject. This is what a correctly specified null looks like when it is true, and it is more informative than either group appearing alone.

Most apparent performance was exposure. The highest Sharpe ratios in the study belong to a forecaster that never changes its mind. An evaluation omitting a buy-and-hold reference would have credited a drift model with skill it does not possess. This is a mundane failure mode and, we suspect, a common one.

The result is consistent with weak-form efficiency at daily frequency for large-capitalisation cryptocurrencies over this period, net of realistic costs [27]. It does not establish that no cryptocurrency signal exists. It establishes that this feature set, at these horizons, on these assets, does not produce one that survives its own error bars.

7 Limitations

Asset selection is not survivorship-free, as noted in Section 3. We use daily spot bars only, with no order book, funding rates, intraday structure, or cross-exchange information; short-horizon predictability, if it exists, is more likely to live at frequencies this data cannot resolve. The feature set is compact and the model families are three, so richer features or sequence models might change the picture, though the leakage discipline would not need to change with them. Short positions are modelled as fully collateralised, returning $-r$ with no borrow cost, margin requirement, or liquidation, which is reasonable for assessing signal quality and optimistic as an execution model. Costs are a flat per-side estimate that does not scale with size or volatility. Intervals are percentile bootstrap rather than BCa, adequate for a reject/do-not-reject reading and mildly biased for a statistic as asymmetric as the Sharpe ratio. The deflated Sharpe ratio understates the true search, deflating for the models raced within one asset–horizon rather than for the whole grid and for the design choices fixed before the grid was run. Finally, the sample end date is unpinned, so re-running extends the window; the evaluated window is stamped into the generated report so that any table can be tied to the data producing it.

8 Conclusion

Across 18 model, asset, and horizon combinations evaluated by purged walk-forward cross-validation from 2020 to 2026, we find a faint statistical signal in BTC at the one-day horizon that survives false-discovery control and does not survive family-wise control. It does not convert into a forecast beating an ex-ante drift estimate, into sign-timing skill, or into a Sharpe ratio distinguishable from zero after costs. The strategies with the highest Sharpe ratios in the study are long-only exposure.

The negative result is the deliverable, and it is reported alongside the tests that would have detected a positive one and the corrections that would have removed a spurious one.

Reproducibility. All results, figures, and tables in this paper are generated by a single command from a public repository, with a test suite of 154 tests covering the leakage guarantees, the statistical formulas against hand-computed reference values, and the simulation demonstrating the Diebold–Mariano bias. The tables reproduced here are emitted directly from the committed results file rather than transcribed.

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A Full results

Table 3 gives forecast accuracy and all three predictive-accuracy tests for every model, asset, and horizon. Table 4 gives net-of-cost performance with bootstrap intervals, the phase spread, and both Sharpe corrections. Entries in bold are significant at the uncorrected 5% level; Section 5.2 gives the corrected counts, which are what should be read.

Table 3: Forecast accuracy and predictive-accuracy tests, all settings. Each test is reported as statistic (p). DM negative favours the model, CW positive favours the model, PT positive indicates sign-timing skill.

Model	RMSE	R^2_{OS}	Dir. acc.	Rank IC	DM (p)	CW (p)	PT (p)
<i>BTC, h = 1</i>							
Random walk	0.0301	0.0005	—	—	—	—	—
Historical mean	0.0301	0.0000	0.506	0.002	+0.18 (0.860)	+1.03 (0.152)	—
AR(1)	0.0301	-0.0013	0.513	0.033	+0.34 (0.737)	+1.24 (0.107)	+1.24 (0.214)
Ridge	0.0302	-0.0079	0.506	0.049	+0.76 (0.447)	+2.62 (0.004)	+0.51 (0.609)
Elastic net	0.0301	0.0031	0.505	0.046	-0.46 (0.644)	+2.57 (0.005)	+0.25 (0.806)
GBM	0.0301	0.0007	0.501	0.019	-0.04 (0.970)	+2.02 (0.022)	-0.38 (0.704)
<i>BTC, h = 7</i>							
Random walk	0.0804	0.0072	—	—	—	—	—
Historical mean	0.0807	0.0000	0.529	-0.018	+0.41 (0.685)	+1.01 (0.156)	—
AR(1)	0.0807	-0.0002	0.529	-0.023	+0.42 (0.675)	+1.00 (0.159)	—
Ridge	0.0825	-0.0466	0.523	0.044	+1.85 (0.064)	+1.64 (0.050)	+1.41 (0.159)
Elastic net	0.0817	-0.0258	0.523	0.043	+1.43 (0.152)	+1.76 (0.039)	+1.24 (0.217)
GBM	0.0809	-0.0055	0.528	0.032	+0.57 (0.569)	+1.25 (0.105)	+0.74 (0.459)
<i>ETH, h = 1</i>							
Random walk	0.0407	0.0006	—	—	—	—	—
Historical mean	0.0407	0.0000	0.515	0.019	+0.25 (0.806)	+0.81 (0.209)	—
AR(1)	0.0408	-0.0020	0.517	0.036	+0.47 (0.639)	+1.16 (0.124)	+0.96 (0.340)
Ridge	0.0411	-0.0167	0.509	0.053	+1.42 (0.157)	+1.86 (0.031)	+0.54 (0.588)
Elastic net	0.0409	-0.0051	0.515	0.040	+0.82 (0.413)	+1.37 (0.085)	+1.02 (0.308)
GBM	0.0408	-0.0009	0.512	0.021	+0.29 (0.775)	+1.42 (0.078)	-0.30 (0.766)
<i>ETH, h = 7</i>							
Random walk	0.1073	0.0057	—	—	—	—	—
Historical mean	0.1076	0.0000	0.514	0.039	+0.39 (0.697)	+0.89 (0.188)	—
AR(1)	0.1076	-0.0002	0.514	0.033	+0.40 (0.686)	+0.87 (0.191)	—
Ridge	0.1123	-0.0908	0.476	-0.047	+3.15 (0.002)	-0.79 (0.785)	-3.29 (0.001)
Elastic net	0.1115	-0.0749	0.479	-0.052	+2.98 (0.003)	-0.98 (0.837)	-3.33 (0.001)
GBM	0.1082	-0.0126	0.512	-0.042	+1.19 (0.235)	+0.55 (0.291)	+0.21 (0.835)
<i>SOL, h = 1</i>							
Random walk	0.0498	0.0153	—	—	—	—	—
Historical mean	0.0502	0.0000	0.481	-0.029	+3.01 (0.003)	-1.23 (0.891)	—
AR(1)	0.0504	-0.0062	0.478	-0.007	+2.39 (0.017)	-0.77 (0.779)	-0.81 (0.415)
Ridge	0.0509	-0.0262	0.500	0.010	+2.10 (0.036)	+0.31 (0.379)	+0.03 (0.977)
Elastic net	0.0504	-0.0080	0.491	0.013	+1.56 (0.119)	+0.15 (0.441)	-0.64 (0.522)
GBM	0.0505	-0.0095	0.492	0.023	+2.00 (0.046)	+0.82 (0.207)	+0.56 (0.576)
<i>SOL, h = 7</i>							
Random walk	0.1345	0.1023	—	—	—	—	—
Historical mean	0.1420	0.0000	0.467	-0.103	+3.63 (0.000)	-1.74 (0.959)	—
AR(1)	0.1419	0.0007	0.467	-0.098	+3.62 (0.000)	-1.71 (0.957)	—
Ridge	0.1449	-0.0416	0.476	-0.039	+3.22 (0.001)	-1.43 (0.923)	-1.73 (0.083)
Elastic net	0.1435	-0.0210	0.491	-0.034	+3.19 (0.001)	-1.41 (0.921)	-0.45 (0.649)
GBM	0.1446	-0.0379	0.471	-0.080	+3.32 (0.001)	-1.39 (0.918)	-0.49 (0.625)

Table 4: Net-of-cost performance. Sharpe ratios annualise at $365/h$. Phases gives the range of net Sharpe across all h start offsets of the trading schedule. DSR is an upper bound, as discussed in Section 7.

Model	Sharpe	95% CI	Phases	Max DD	PSR	DSR
<i>BTC, h = 1</i>						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.84	[0.01, 1.61]	–	-76.6%	0.98	0.87
AR(1)	0.21	[-0.51, 0.90]	–	-82.7%	0.70	0.34
Ridge	0.51	[-0.18, 1.23]	–	-65.7%	0.89	0.63
Elastic net	0.33	[-0.35, 1.02]	–	-74.3%	0.79	0.45
GBM	0.22	[-0.58, 1.02]	–	-89.4%	0.71	0.35
Buy & hold	0.84	[0.01, 1.61]	–	-76.6%	0.98	–
<i>BTC, h = 7</i>						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
AR(1)	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
Ridge	0.57	[-0.27, 1.44]	[0.37, 0.80]	-78.5%	0.92	0.65
Elastic net	0.51	[-0.54, 1.55]	[0.35, 0.90]	-86.1%	0.90	0.60
GBM	0.82	[-0.08, 1.69]	[0.50, 0.82]	-54.6%	0.98	0.85
Buy & hold	0.80	[-0.12, 1.63]	–	-75.9%	0.98	–
<i>ETH, h = 1</i>						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.80	[-0.02, 1.57]	–	-79.4%	0.97	0.64
AR(1)	-0.15	[-0.96, 0.57]	–	-93.7%	0.36	0.02
Ridge	-0.18	[-0.90, 0.61]	–	-97.5%	0.33	0.02
Elastic net	-0.11	[-0.84, 0.64]	–	-97.7%	0.39	0.03
GBM	0.91	[0.00, 1.74]	–	-80.4%	0.99	0.74
Buy & hold	0.80	[-0.02, 1.57]	–	-79.4%	0.97	–
<i>ETH, h = 7</i>						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
AR(1)	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
Ridge	-0.11	[-0.98, 0.81]	[-0.35, -0.11]	-99.2%	0.40	0.03
Elastic net	-0.13	[-0.95, 0.72]	[-0.40, -0.07]	-98.9%	0.38	0.03
GBM	0.92	[0.02, 1.80]	[0.49, 1.05]	-84.5%	0.99	0.76
Buy & hold	0.76	[-0.09, 1.56]	–	-77.3%	0.97	–
<i>SOL, h = 1</i>						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.25	[-0.63, 1.13]	–	-96.3%	0.71	0.43
AR(1)	-0.19	[-1.01, 0.67]	–	-97.5%	0.34	0.13
Ridge	0.05	[-0.89, 0.82]	–	-97.4%	0.54	0.27
Elastic net	-0.22	[-1.21, 0.63]	–	-97.1%	0.32	0.12
GBM	0.43	[-0.49, 1.27]	–	-91.3%	0.83	0.59
Buy & hold	0.25	[-0.63, 1.13]	–	-96.3%	0.71	–
<i>SOL, h = 7</i>						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.23	[-0.75, 1.09]	[0.20, 0.25]	-96.2%	0.69	0.47
AR(1)	0.23	[-0.75, 1.09]	[0.20, 0.25]	-96.2%	0.69	0.47
Ridge	0.45	[-0.43, 1.23]	[-0.83, 0.57]	-89.7%	0.84	0.66
Elastic net	0.60	[-0.19, 1.35]	[-0.55, 0.60]	-83.9%	0.91	0.77
GBM	0.22	[-0.62, 0.94]	[0.09, 0.22]	-94.9%	0.69	0.46
Buy & hold	0.23	[-0.75, 1.09]	–	-96.2%	0.69	–

B Equity curves

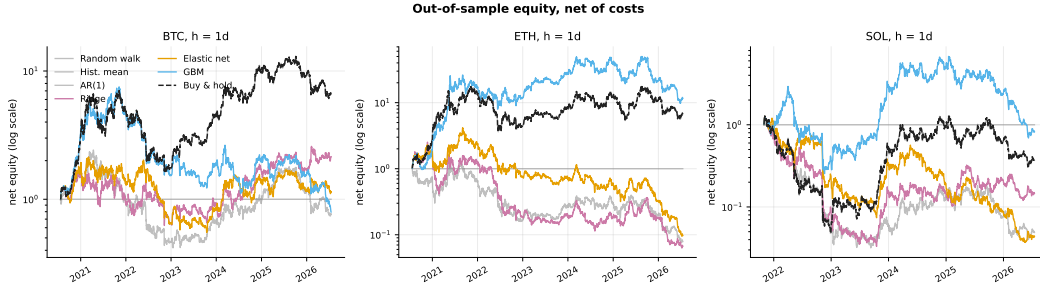


Figure 8: Out-of-sample equity net of costs at $h = 1$, against the always-long reference on the same schedule and cost model. Starting capital is unity and the vertical scale is logarithmic.